Samuel Adole

Leeds, UK | Open to remote / hybrid | Full right to work in the UK | adoleichasamuel@gmail.com | +44 7757 621 094 | linkedin.com/in/samuel-adole | samueladole.github.io

PROFESSIONAL PROFILE

- Senior Software Engineer focused on the engineering layer beneath AI products, with six years' experience architecting and owning production backend systems across healthcare, fintech, and high-throughput SaaS. Currently building a real-time observability and evaluation platform for RAG systems as an MSc Data Science dissertation, applying production tracing and monitoring practice to LLM pipelines. Deep expertise in Python (Django, Django REST Framework, FastAPI) and TypeScript/Node.js, with a record of taking distributed services from architectural design to production on AWS, including a fintech platform architected from scratch as sole backend lead. Leads technical direction, owns system design decisions, and mentors junior engineers.

KEY ACHIEVEMENTS

- Cut backend API latency by 30% and lifted data-API throughput by 40% on production systems serving distributed, multi-venue and multi-tenant workloads.
- Architected two greenfield backend platforms end-to-end (fintech and supply-chain SaaS), acting as sole backend lead on the fintech build.
- Set the engineering quality bar across a microservices estate through code review, CI/CD gates, and 80%+ test coverage, mentoring engineers on testing and observability practices.

SKILLS

AI & LLM Engineering: Anthropic Claude API, OpenAI API, LangChain, Pydantic AI, LLM application design, prompt design, tool/function calling, structured and schema-validated output, retrieval-augmented generation (RAG), embeddings and vector retrieval
LLM Evaluation & Observability: RAGAS, RAG evaluation (retrieval relevance, groundedness, answer quality), custom LLM pipeline tracing and instrumentation, ChromaDB, BM25, Reciprocal Rank Fusion, real-time metrics and monitoring
Machine Learning & Data: PyTorch, TensorFlow, STT/ASR training pipelines, dataset preparation and augmentation, ETL, statistical modelling, NLP
Languages: Python, TypeScript, JavaScript (ES6+), SQL, Bash
System Architecture & Design: Distributed systems design, microservices decomposition, event-driven architecture, API and contract design, multi-tenant SaaS, high availability and horizontal scaling, caching and performance strategy, fault tolerance, technical design documentation
Backend & APIs: Django, Django REST Framework, FastAPI, Flask, Falcon, Node.js, NestJS, Express, REST, WebSockets, OpenAPI/Swagger
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes & Helm, Ansible, Linux, GitHub Actions CI/CD, infrastructure-as-code
Data & Storage: PostgreSQL, Redis, MongoDB, Celery, schema design, query optimisation, ETL pipelines, data reconciliation
Testing & Quality: PyTest, Jest, React Testing Library, TDD, code review, CI/CD quality gates, structured logging, observability
Security & Access: JWT, RBAC, secure REST API design, audit trails, healthcare compliance and data-consistency controls
Frontend: React, Next.js, Redux Toolkit, TypeScript, component-driven UI, real-time and streaming interfaces
Leadership & Professional: System design ownership, engineering standards, mentoring and coaching, code review, Agile/Scrum, technical documentation, stakeholder communication, incident response, remote-first collaboration

PROJECTS

Real-Time Observability & Evaluation Platform for RAG Systems In progress – 2026
MSc Data Science dissertation — Leeds Beckett University

- Built a framework-agnostic RAG platform in Python 3.12 using a dual-index retrieval layer: ChromaDB (HNSW, cosine, 384-dim all-MiniLM-L6-v2 embeddings) fused with BM25Okapi via a custom Reciprocal Rank Fusion layer (k=60, 0.6 dense / 0.4 sparse) without pulling in a retrieval framework.
- Implemented automated quality evaluation using RAGAS 0.4 (context relevance, faithfulness, answer correctness) with Qwen2.5 7B as a fixed third-model-family judge; surfaced as live per-query metrics rather than batch-scored after the fact.
- Evaluated LangSmith, Arize Phoenix, TruLens, and OpenTelemetry LLM extensions before building custom instrumentation: a time.perf_counter() Timer across four stages (embed, retrieve, generate, evaluate), writing one orjson document per query with token counts reconciled against tiktoken. This decomposition revealed RAGAS evaluation accounts for 85.7% of end-to-end latency.

EXPERIENCE

Senior Software Engineer Apr 2022 – Present
Code & Care *Remote, UK*

- Software engineering consultancy — engineers embedded within client product teams

- Client: TOCA Social — London, UK (Oct 2023 – Nov 2025)
- Led design and delivery of public REST and WebSocket APIs (TypeScript / Node.js) powering real-time, multi-venue synchronisation across a distributed microservices estate on AWS, handling up to 10,000 concurrent requests per second, 5,000–10,000 events per day with weekend peaks, and AWS service SLAs of 99.5% to 99.999% uptime (S3, CloudWatch, EC2, Lambda).
- Owned the TOCA Connects integration layer end-to-end: a cross-studio data-synchronisation service connecting internal platform services to third-party venue-management systems, defining the reconciliation and consistency guarantees enterprise integrations demand.
- Reduced backend API latency by 30% through profiling, query optimisation, and caching; introduced structured logging and monitoring that measurably shortened mean time to detection on production incidents.
- Set and upheld engineering standards through code review and CI/CD quality gates (GitHub Actions), sustaining 80%+ automated test coverage (Jest, React Testing Library) and preventing regressions at scale.
- Client: Fulton Market Group (FMG) — Australia (Apr 2022 – Aug 2023)
- Architected and built high-throughput internal REST APIs in Python (Falcon) for a multi-tenant supply-chain SaaS platform, with strict validation, structured error handling, and full OpenAPI documentation.
- Delivered real-time operational dashboards (WebSocket-driven Python services, React), giving operations teams live visibility of 20,000+ shipment events per month.
- Introduced structured logging, polling fallbacks, and fault-tolerance patterns across distributed services, significantly reducing on-call incident volume.

Full-Stack Software Engineer Nov 2021 – Mar 2022
Intron Health *California, US — Remote*

- Healthcare technology — EMR integration under compliance constraints
- Built ETL pipelines integrating AI transcription output into downstream EMR systems via structured PostgreSQL schemas and secure REST APIs, meeting accuracy, consistency, and audit-trail requirements under healthcare compliance.
- Developed preprocessing and data-augmentation tooling for STT/ASR model training (PyTorch, TensorFlow), covering 300+ accent profiles with measurable gains in accuracy and pipeline repeatability.
- Authored integration specifications covering data flow, failure modes, and rollback behaviour, giving a distributed team a dependable reference for asynchronous review across time zones.

Full-Stack Software Engineer Aug 2021 – Dec 2021
Amenta Media AB (Likvio) *Gothenburg, Sweden — Remote*

- Fintech SaaS — greenfield Django / FastAPI platform build
- Architected the Likvio backend platform from scratch as sole backend lead, defining the data model, service boundaries, and integration architecture, and delivered it to production on FastAPI and Django REST Framework.
- Integrated the Fortnox financial API to automate contract lifecycle and invoice synchronisation, and managed high-volume asynchronous processing (Celery, PostgreSQL) with retry and reconciliation logic holding 100% data integrity across concurrent financial operations.
- Onboarded and mentored junior developers, establishing coding standards, code-review practice, and OpenAPI documentation that brought them and external integrators to independent contribution.

Software Engineer Jan 2021 – Apr 2021
V3C Technologies *Abuja, Nigeria*

- Multi-role education platform
- Designed Flask REST APIs with JWT authentication and RBAC across teacher, parent, and student portals; standardised OpenAPI documentation and drove architectural improvements that cut developer integration time by 30%.

Full-Stack Python Developer Dec 2019 – Jan 2021
eHealth4everyone *Abuja, Nigeria*

- Health-data / DHIS2 platform — regulated public-health domain
- Built and operated a DHIS2 hosting platform (Django, Ansible), automating provisioning and configuration as infrastructure-as-code in a public-health data context.
- Designed high-availability data APIs (Django REST Framework, Redis, AWS EC2/RDS) for the HealthThink dataset; caching and auto-scaling improved throughput by 40%.

CERTIFICATIONS

- AWS Machine Learning Foundations — Amazon Web Services

EDUCATION

Leeds Beckett University
MSc Data Science. *Sep 2026 (expected)*
Bingham University Karu
BSc Computer Science. *Sep 2018*